

MLFF PERF-P2 Cloud CPU Qualification

Lazy TARGET-DATA2C v2 versus exhaustive v1

mdstats project

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Scope

PERF-P2 is benchmarked against the exhaustive TARGET-DATA2C v1 oracle using fresh processes per timing sample. The source is the complete PERF-P0 native 37,633-frame target coverage reference. A deterministic frame-index remap aligns the reference with the DATA2C role-order contract.

The host is an Intel Xeon Platinum 8573C with nine visible logical CPUs, an 8-core cgroup CPU quota, and a 4 GiB cgroup memory limit.

Scientific equivalence

Two supplied-data-derived cases were tested.

Case	v1 Stage-A survivors	v2 Stage-A survivors	Survivor evidence
Exhaustive fallback	2048, 4096, 8192	2048, 4096, 8192	exact
Forced early stop	128, 256, 512, 1024	128, 256, 512, 1024	exact

The forced early-stop fixture changes local radii to a deliberately permissive value and removes extent channels only to exercise the safe stop branch. It is not a production coverage-policy change.

Worker counts 1 and 4 produce the same v2 content digest and scientific signature.

Scientific benchmark digest:

ae55c560995791174ac63e2d894ec685d74a02c389eb0a955d87e77cfd9f18f9.

Fresh-process CPU measurements

Forced early stop

Metric	Exhaustive v1	Lazy v2	Change
Median wall	7.867 s	1.556 s	-80.23%
Wall range	7.602–8.898 s	1.505–1.713 s	non-overlapping
Median peak RSS	327.07 MiB	316.23 MiB	-3.31%
Master-order entries	8192	1024	8x fewer
Serialized authority	4,729,481 B	591,058 B	-87.50%

v2 stops after materializing 128, 256, 512, and 1024. Rungs 2048, 4096, and 8192 are explicitly recorded as intentionally unmaterialized.

Exhaustive fallback

v2 materializes all configured rungs because only three qualify. The observed fresh-process wall ranges overlap and are scheduler-sensitive. The recorded three-sample medians are 11.003 s for v1 and 8.726 s for v2, but that apparent median difference is not promoted as a portable performance claim.

Interpretation

PERF-P2 removes exact work only after the Stage-A survivor set is already fixed. The strong early-stop timing separation and large reduction in retained authority size are therefore the relevant performance evidence. The fallback case establishes that v2 remains scientifically exhaustive when the stop predicate is not reached.

No GPU or MACE training timing is part of this gate.